Curriculum Vitae

PERSONAL INFORMATION

Pablo Pareja Tobes

Pintor Lopez Mezquita 5 5C, Granada, 18002, Spain

+34 656257101

pabloparejatobes@gmail.com

http://www.pablo-pareja.com

GitHub user: pablopareja

Sex Male | Date of birth 20/11/1985 | Nationality Spanish

POSITION

Bioinformatics IT consultant/developer/researcher at *Oh no sequences!* *Era7 Bioinformatics*

WORK EXPERIENCE

August 2007 to present

Bioinformatics IT consultant / developer / researcher

Era7 Bioinformatics

- Project Manager
- Project Leader
- Data Scientist
- Full Stack developer
- Data Analyst
- Data visualization expert

Direction and development of several kinds of projects including: Web applications *(Full Stack),* applications based on cloud-computing, databases, implementation of algorithms, pipelines and bioinformatics methods.

Creator of **Bio4j**, creator and leader of **Biographika** project and main developer of many other tools and open source projects such as **BG7** or **MG7**.

Consultant activity for company management.

April 2012 – June 2012

Visiting scholar

The Ohio State University, United States

Using the graph database paradigm to improve GRASSIUS database plus its integration with Bio4j platform

March 2006 – June 2006

Research fellowship

Models of Decision and Optimization (MODO) Research Group (University of Granada)

Technical assistance in GUI design, web development, assistance in the organization of Workshop NICSO

EDUCATION AND TRAINING

September 2010 – June 2011

Bachelor's degree in Translation and Interpreting

University of Granada, Spain

'A' language: **English**

'B' language: **French**

October 2006 – July 2007

Erasmus exchange year on Computer Science

École polytechnique fédérale de Lausanne (EPFL), Switzerland

1st year of Master's on Computer Science

September 2003 – June 2006

Bachelor's degree on Computer Science

University of Granada, Spain

Data structures, Software engineering, Operating systems, Computational models, Relational databases, Algorithms, Artificial Intelligence, Algebra, Physics, Electronics and Computer Technology among others.

PERSONAL SKILLS

Mother tongue(s) Spanish

Other language(s)

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C2	C2	C2	C2	C2
	Cambridge certificate in Advanced English C1				
French	C1	C2	C1	C1	C1
Italian	C2	C2	C2	C2	C1
Portuguese	A2	A2	A2	A2	A2
German	A1	A2	A2	A1	A1

Levels: A1/2: Basic user - B1/2: Independent user - C1/2 Proficient user
Common European Framework of Reference for Languages

Personal skills

- Innovative
- Fast learner
- Self motivated
- Productive
- Eager to learn new technologies and paradigms
- Communication skills
- Analytical thinking
- Team working skills
- Complex problem solving
- Creative
- Passionate about new challenges
- Curious
- Energetic

TECHNICAL SKILLS

- **Programming languages:**

Proficient: *Java*, *Javascript, Actionscript, Flex, Flash*
Familiar with*: C, C++*

- **Software engineering:**

Complex problem solving, Active Learning, Quality control analysis, Time management, Systems evaluation, Self-directed learning skills, Focus on usable an mantainable code, Large scale data problem solving

- **Databases:**

Database design, tuning, performance analysis, optimization…

Graph/NoSQL databases:
Expert: Neo4j, Titan
Familiar with: OrientDB, MongoDB, Berkeley DBXML, Berkeley DB

Relational Databases:
SQL query design, analysis, performance tuning...
Proficient: PostgreSQL, MySQL, SQL

- **Data visualization:**

D3.js, Sigma.js, X3DOM, Crossfilter.js, Gephi, Cytoscape, Bootstrap

- **Cloud Computing:**

Amazon Web Services: *SQS, SNS, EC2, S3, AWS Java SDK, AWS CLI*
Experience developing distributed systems, dealing with asynchronous architectures…

- **Open source development:**

Leader of open source projects, open source advocate, participant as organization leader and mentor at the Google Summer of Code 2014 with Bio4j project.

- **GitHub**

- **Scalability**

- **Big Data**

- **Full stack web application developer**

- **Algorithm design**

- **Data mining**

- **Experience dealing with many file formats including:**
XML, JSON, GraphML, GEXF, GraphSON among others.

Other skills

I'm a semi-professional musician (singer and pianist) as well as amateur composer..
In terms of official education, I passed the 4 th year piano exam at the Conservatory of Granada.

Driving licence

B

Publications

Important biological information uncovered in previously unaligned reads from chromatin immunoprecipitation experiments (ChIP-Seq)
Wilberforce Zachary Ouma, Maria Katherine Mejia-Guerra, Alper Yilmaz, Pablo Pareja-Tobes, We Li, Andrea I, Erich Grotewold.
Scientific Reports 03/2015; 5. DOI: 10.1038/srep08635 **5.58 Impact Factor**

Metagenomic Analysis of Milk of Healthy and Mastitis-Suffering Women
Esther Jiménez, Javier de Andrés, Marina Manrique, Pablo Pareja-Tobes, Raquel Tobes, Juan F Martínez-Blanch, Francisco M Codoñer, Daniel Ramón.
Journal of Human Lactation 05/2015; 31(3). DOI: 10.1177/0890334415585078 **1.99 Impact Factor**

Gene Calling and Bacterial Genome Annotation with BG7
Raquel Tobes, Pablo Pareja-Tobes, Marina Manrique, Eduardo Pareja-Tobes, Evdokim Kovach, Alexey Alekhin, Eduardo Pareja.
Methods in Molecular Biology 01/2015; 1231:177-89. DOI:10.1007/978-1-4939-1720-4_12 **1.29 Impact Factor**

BG7: A New Approach for Bacterial Genome Annotation Designed for Next Generation Sequencing Data.
Pablo Pareja-Tobes, Marina Manrique, Eduardo Pareja-Tobes, Eduardo Pareja, Raquel Tobes
PLoS ONE 11/2012; 7(11):e49239. DOI: 10.1371/journal.pone.0049239 **3.23 Impact Factor**

Escherichia coli EHEC Germany outbreak preliminary functional annotation using BG7 system
Raquel Tobes, Marina Manrique, Pablo Pareja-Tobes, Eduardo Pareja-Tobes, Eduardo Pareja
Nature Precedings 06/2011; **DOI:** 10.1038/npre.2011.6001.1

Search spaces representation in optimization problems
P. Pareja-Tobes, David A. Pelta, Alejandro Sancho-Royo, José L. Verdegay
Expert Systems with Applications 05/2008; 34(4):2891-2895. DOI: 10.1016/j.eswa.2007.05.046 **2.24 Impact Factor**

ExtraTrain: a database of Extragenic regions and Transcriptional information in prokaryotic organisms
Eduardo Pareja, Pablo Pareja-Tobes, Marina Manrique, Eduardo Pareja-Tobes, Javier Bonal, Raquel Tobes.
BMC Microbiology 02/2006; 6(1):29. DOI: 10.1186/1471-2180-6-29 **2.73 Impact Factor**

Projects

Bio4j

Bio4j is an **open source bioinformatics graph data platform**, integrating most data available in Uniprot KB (SwissProt + Trembl), Gene Ontology (GO), UniRef(50,90,100), NCBI Taxonomy, and Expasy Enzyme DB.

Bio4j provides a completely new and powerful framework for protein related information querying and management.
Bio4j was selected as an organization for the Google Summer of Code in 2014
Online preprint *(pending submission)*

Biographika

Rich interactive data visualizations on the web for the research community
Biographika online preprint *(pending submission)*

BG7

An open source bacterial genome annotation system, specifically designed for NGS data.

Pesentations/Seminars

Bio4j: bigger, faster, leaner
FOSDEM 2014, Brussels, Belgium

Graph Databases in Life Sciences Workshop, 2012 Ghent, Belgium
Bio4j: a pioneer graph based database for the integration of biological Big Data

BioMicroMeeting: BIG meeting MIT, 2011 Cambridge MA, USA
Bio4j Graph based analysis and visualization

UP 2011 Cloud Computing Conference *(December 2011)*
Bio4j, the bioinformatics graph database on the cloud.

Conference Proceedings

BG7, A new system for bacterial genome annotation designed for NGS data.
Marina Manrique, Pablo Pareja, Eduardo Pareja-Tobes, Eduardo Pareja, Raquel Tobes
Applied Bioinformatics & Public Health Microbiology 2011,1-3 June 2011, Wellcome Trust Genome Campus, Hinxton, Cambridge, UK.

Cloud computing and NGS: massively parallel computing for massively parallel sequencing
Pareja-Tobes, E; Manrique, M; Pareja-Tobes, P; Pareja, E; Tobes, R;
NGS2009: Conference on Next Generation Sequencing: Challenges and Opportunities. Barcelona (Spain), 1-3rd October 2009

GS-Palinsight: Genome scale palindromicity analyzer tool.
Aranda-Olmedo I, Manrique M, Espín V, Pareja-Tobes P, Pareja-Tobes E, Pareja E, Tobes R.
Meeting of the National Network of Systems Biology. Murcia Nov. 2007.

KEYPROTEINS: A database of sequences of Mammalian Transmembrane Proteins.
Manrique M, Tobes R, Pareja-Tobes P and Pareja E
International Workshop Integrative Bioinformatics 4th annual meeting, 10th to 12th September 2007 University of Ghent, Belgium

ExtraTrain: a database of Extragenic regions and Transcriptional information in prokaryoticorganisms.
Pareja E, Pareja-Tobes P, Manrique M, Pareja-Tobes E, Bonal J and Tobes R
.2nd FEMS Congress of European Microbiologists. Madrid July 2006.